

claude-windows / 8d4866bf-8edd-4da9-909e-022dde12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022dde12675\subagents\workflows\wf_3d035317-127\agent-a208f32362920793f.jsonl`  
- SHA-256: `8213a54c76d89362e2b4c1da1d7b6a8040fbba3226fce97e25f4c41ece4787a0`  
- Source modified: `2026-06-22T09:47:37+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_3d035317-127`  
- session_id: `8d4866bf-8edd-4da9-909e-022dde12675`
```

Transcript

1. user (2026-06-22T09:46:23.303Z)

You are fixing staleness in ONE doc of the KhelSutra repo, applying an audit that already ran. Be SURGICAL: minimal, targeted edits that match the audited fix; preserve the doc's voice, structure, and markdown formatting. Do NOT wholesale-rewrite a doc.

KhelSutra repo root: C:\Users\avidu\Projects\kheelsutra
ENGINE repo root (sibling, for cross-repo verification): C:\Users\avidu\Projects\badminton-highlight-indexer (origin/master freshly fetched)
Today: 2026-06-22. KhelSutra is on branch docs/freshness-sweep (off master).

YOUR DOC (edit ONLY this file): C:\Users\avidu\Projects\kheelsutra\deploy\DEPLOY_ALPHA.md

The audit findings are in this JSON file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022dde12675\tasks\w3dbzrpby.output
Shape: { result: { perDoc: [{doc, overallStatus, findings, summary}], staleFindings: [{doc, claim, docLocation, verdict, evidence, suggestedFix, severity}] } }.
Read it and select ONLY entries where doc === "deploy/DEPLOY_ALPHA.md". Your worklist = that doc's staleFindings. Also skim that doc's perDoc.findings for "unverifiable" items ? do NOT act on those unless you can now positively verify them.

For each stale finding:

1. RE-VERIFY it yourself against the LIVE repo before editing ? Read the doc lines + the cited code. For any claim about the ENGINE (paths like backend/main.py, /api/*, native_wasb_runner.py, auth/CORS), verify against the engine repo's origin/master: e.g. ``git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "<symbol>" origin/master -- backend/`` or ``git -C ... show origin/master:backend/main.py``. Trust nothing blindly ? the audit itself may be slightly off and engine line numbers DRIFT between PRs.
2. If confirmed, apply the suggestedFix as a SURGICAL Edit. For brittle ENGINE line-number anchors, prefer SYMBOL-based references (e.g. ``backend/main.py::get_config``) over chasing exact line numbers ? but only after confirming the symbol exists on engine origin/master.
3. DOCS ONLY: if a fix would require a CODE change (e.g. a comment in site/server.js), do NOT edit code ? record it under codeChangesNeeded.

Hard rules:

- Edit ONLY your assigned doc file. Run NO git WRITE commands (no add/commit/checkout/branch/stash) ? read-only git for verification only.
- Keep status markers HONEST: flip Todo->Done only WITH the PR number / file evidence; never invent a completion.
- If a finding is wrong, not reproducible, or too risky to auto-apply, SKIP it and explain under skipped.
- Do not "improve" beyond the findings; no scope creep.

Return the structured report.

2. assistant (2026-06-22T09:46:25.873Z)

I'll start by reading the audit findings and the target doc.

3. user (2026-06-22T09:46:28.307Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-591 of 1722 total (61859 tokens, cap 25000). Call Read with offset=592 limit=591 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      {
2      "summary": "Audit every KhelSutra doc for staleness against the live repo, returning
structured findings",
3      "agentCount": 15,
4      "logs": [
5      "audited 15/15 docs; 44 stale findings; archive-candidates: none"
6      ],
7      "result": {
8      "perDoc": [
9      {
10     "doc": "NEXT_STEPS.md",
11     "overallStatus": "minor-staleness",
12     "summary": "NEXT_STEPS.md is overwhelmingly accurate against master: routes,
file paths, the Worker/D1 names, the 6-language localization, the 3-step wizard, the per-rally
PR numbers (#11/#14/#21/#28/#30/#31), the appliance/per-rally doc statuses, and every still-
pending \"Queued\" item all check out against code and git. The only real staleness is the \"PRs
through #93 merged\" count, which lags by one because PR #94 (the very commit that wrote
\"through #93\") merged afterward; plus a low-severity GCS-artifact filename mismatch. Not an
archive candidate ? it is the live next-steps tracker.",
13     "findings": [
14     {
15     "claim": "PRs through #93 merged (master).",
16     "docLocation": "NEXT_STEPS.md:13",
17     "verdict": "stale",
18     "evidence": "git log --oneline -1 = '50b2bd7 chore(repo): ... correct stale
NEXT_STEPS claims (#94)'; total 94 commits/PRs on master. Ironically #94 is the commit that
updated this line to '#93', then merged itself, so the doc now lags one PR behind master.",
19     "suggestedFix": "Update to 'PRs through #94 merged (master).'",
20     "severity": "medium"
21     },
22     {
23     "claim": "Where we are (2026-06-20): ... khelsutra.guru is LIVE and fully
shipped (landing + /capture checklist + OG image + segmented Request-Access form).",
24     "docLocation": "NEXT_STEPS.md:5-14",
25     "verdict": "correct",
26     "evidence": "site/server.js:98 maps /capture->capture.html;
site/public/og.png + capture.html tracked; index.html:191-193 textareas + lines 537-581
academy/individual/gpu/storage fields. Header date 2026-06-20 is slightly behind the latest work
(PR #94 merged 2026-06-22) but the substance holds.",
27     "suggestedFix": "Optionally bump the 'Where we are' date to 2026-06-22 to
match the latest merged work.",
28     "severity": "low"
29     },
30     {
31     "claim": "On Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-
access): the form persists to D1 and emails a summary via an Email Routing send binding.",
32     "docLocation": "NEXT_STEPS.md:7-9",
33     "verdict": "correct",
34     "evidence": "site/wrangler.toml: name = \"hidden-king-khelsutra\";
[[d1_databases]] database_name = \"khelsutra-access\"; [[send_email]] name = \"NOTIFY\".
site/src/index.js:80 handles POST /api/request-access -> insert into D1 binding DB.",
35     "suggestedFix": "",
36     "severity": "low"
37     },
38     ]
39     },
40     }
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38         {
39             "claim": "CI gates every PR ? three jobs: full site npm test, web/ SPA build
+ coverage floor, and a real-browser layout check.",
40             "docLocation": "NEXT_STEPS.md:9-12",
41             "verdict": "correct",
42             "evidence": "site/package.json scripts include test (node --test),
test:layout (layout.browser.mjs); web has a coverage floor (#80). PR #94 message confirms 'three
jobs: site npm test = 86 green, web SPA build+coverage, real-browser layout'. Non-browser site
suite ran green (61 tests in the 4 non-layout files).",
43             "suggestedFix": "",
44             "severity": "low"
45         },
46         {
47             "claim": "The form also runs on a dependency-free Node + SQLite host
(site/server.js), live on the DO droplet -> http://168.144.126.176:8787 (systemd khelsutra-
site).",
48             "docLocation": "NEXT_STEPS.md:11-13",
49             "verdict": "correct",
50             "evidence": "site/server.js exists, uses node:sqlite (header comment line
2). IP 168.144.126.176 + port 8787 + 'khelsutra-site' all appear in
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48 ('/srv/khelsutra-site, node :8787') and deploy refs.
Live-status is external/unverifiable from the repo, but the host code and identifiers are
correct.",
51             "suggestedFix": "",
52             "severity": "low"
53         },
54         {
55             "claim": "Docs: docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md.",
56             "docLocation": "NEXT_STEPS.md:13-14",
57             "verdict": "correct",
58             "evidence": "git ls-files: both docs/REQUEST_ACCESS_FORM.md and
docs/PORTABILITY.md are tracked.",
59             "suggestedFix": "",
60             "severity": "low"
61         },
62         {
63             "claim": "DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22): localized to
all 6 languages (en/kn/hi/es/da/id); full record docs/archives/FRIENDLY_AND_MULTILINGUAL.md.",
64             "docLocation": "NEXT_STEPS.md:17-21",
65             "verdict": "correct",
66             "evidence": "docs/archives/FRIENDLY_AND_MULTILINGUAL.md is tracked; git log
shows PR #93 'docs: archive the Friendly & Multilingual tracker'. web/src/i18n/index.ts:25
SUPPORTED_LOCALES = [en,kn,hi,es,da,id]; PR #87 added es/da/id to the marketing site.",
67             "suggestedFix": "",
68             "severity": "low"
69         },
70         {
71             "claim": "Interactive per-rally selection MVP/M0 COMPLETE; Done: P0 design
(#11), P1 scaffold (#14), P2 grid+hook (#21), P3 parseQuery query bar (#28), P4 SelectionFooter
(#30), P5 real load+download (#31). Tracked doc docs/PER_RALLY_SELECTION.md (IN PROGRESS).",
72             "docLocation": "NEXT_STEPS.md:32-41",
73             "verdict": "correct",
74             "evidence": "git log: cb74ce5 #11 'tracked design ... RallyPicker'; aac7e0c
#14 'P1 scaffold'; 62598e7 #21 'P2 selectable rally grid + hook'; d537d19 #28 'P3 deterministic
parseQuery() query bar'; aff8de8 #30 'P4 SelectionFooter + min_shot_count warning'; 8bf200d #31
'P5 load real rallies + Build->compile->preview->download'. docs/PER_RALLY_SELECTION.md status
header = 'IN PROGRESS ? MVP/M0 (P1-P5) COMPLETE'.",
75             "suggestedFix": "",
76             "severity": "low"
77         },
78         {
79             "claim": "P2c first-run setup wizard ? 3-step SetupWizard from
/api/capabilities, 6 languages.",
80             "docLocation": "NEXT_STEPS.md (implied via track) and verified for the
merged #88",

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81         "verdict": "correct",
82         "evidence": "web/src/components/SetupWizard.tsx:22 'const STEPS = 3'; line 1
comment 'Three quick steps'; line 3 'Localized (en/kn/hi/es/da/id)'; web/src/api/client.ts:139
GET /capabilities. git: 7ae9cee #88.",
83         "suggestedFix": "",
84         "severity": "low"
85     },
86     {
87         "claim": "Single-box appliance ? Tracked design
docs/LOCAL_APPLIANCE_ARCHITECTURE.md (DRAFT, owner-gated).",
88         "docLocation": "NEXT_STEPS.md:52-58",
89         "verdict": "correct",
90         "evidence": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md status header: 'Status:
DRAFT ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge)'.",
91         "suggestedFix": "",
92         "severity": "low"
93     },
94     {
95         "claim": "Queued: Always-Use-HTTPS + HSTS (Cloudflare) and Cloudflare Web
Analytics token paste into site/public/index.html ? both still pending.",
96         "docLocation": "NEXT_STEPS.md:44-45",
97         "verdict": "correct",
98         "evidence": "No Strict-Transport-Security/HSTS header found in site/*.js or
wrangler.toml. site/public/index.html:629 still has data-cf-beacon token 'REPLACE_WITH_CF_TOKEN'
(commented-out block). Both correctly listed as not-yet-done.",
99         "suggestedFix": "",
100        "severity": "low"
101    },
102    {
103        "claim": "Droplet -> production: add VM-path SMTP so the Node host also
emails (currently noopNotify).",
104        "docLocation": "NEXT_STEPS.md:46-47",
105        "verdict": "correct",
106        "evidence": "site/server.js:14 imports noopNotify; line 24 'const notify =
noopNotify(); // VM host: no email yet (add SMTP later)'.",
107        "suggestedFix": "",
108        "severity": "low"
109    },
110    {
111        "claim": "Org consolidation DONE; khelsutra (this site repo) transfer is an
owner step still PENDING (unchecked).",
112        "docLocation": "NEXT_STEPS.md:65-73",
113        "verdict": "correct",
114        "evidence": "git remote -v shows origin =
https://github.com/avidullu/khelsutra.git (NOT under Khelsutra org) ? consistent with the '?
owner step pending' marker. Other repos already moved: docs/DRY_RUN_CUJ.md:49 links
github.com/Khelsutra/badminton-highlight-indexer#289.",
115        "suggestedFix": "",
116        "severity": "low"
117    },
118    {
119        "claim": "A real-UI CUJ screencast exists -> promo_cuj.mp4/.gif in
gs://khelsutra-rally-corpus/promo/.",
120        "docLocation": "NEXT_STEPS.md:29-31",
121        "verdict": "unverifiable",
122        "evidence": "No GCS access to confirm the object. The companion doc
docs/DRY_RUN_CUJ.md:85 names the actual artifacts 'khelsutra-bucket-cuj-screencast.mp4' and ':60
khelsutra-i18n-screencast.mp4' in the same gs://khelsutra-rally-corpus/promo/ prefix;
'promo_cuj.mp4' appears nowhere else in the repo, suggesting a possible filename mismatch.",
123        "suggestedFix": "Reconcile the filename with docs/DRY_RUN_CUJ.md (likely
'khelsutra-bucket-cuj-screencast.mp4') or confirm 'promo_cuj.mp4' is a real object in the
bucket.",
124        "severity": "low"
125    },
126    {

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127         "claim": "Control plane (control-plane/) at beta.",
128         "docLocation": "NEXT_STEPS.md:62",
129         "verdict": "correct",
130         "evidence": "control-plane/ directory does not exist in the repo (test -d
looking/aspirational item, not a claim of present existence ? so it is not stale as written.",
131         "suggestedFix": "",
132         "severity": "low"
133     }
134 ]
135 },
136 {
137     "doc": "README.md",
138     "overallStatus": "significant-staleness",
139     "summary": "The repo-structure list, deploy references, and canonical-doc
pointers are all correct, but the README's headline status is badly out of date: it still calls
the repo a \"pre-alpha scaffold\" whose SPA has yet to land \"per milestone M1\", whereas the
web/ SPA is in fact built, multilingual, and shipped (per-rally MVP P0?P5, P2c wizard, in-UI
upload, 6 languages; khelsutra.guru is LIVE). The control-plane/ directory it lists also does
not yet exist.",
140     "findings": [
141     {
142         "claim": "Status: pre-alpha scaffold. SPA lands in `web/` per execution-plan
milestone M1.",
143         "docLocation": "README.md:9",
144         "verdict": "stale",
145         "evidence": "web/src/ has 51 tracked source files incl. web/src/App.tsx;
`git log --oneline -- web/` = 31 commits; merged PRs show the SPA is feature-complete: P2c
first-run wizard (#88), in-UI file upload T4 (#66), per-rally selection MVP P0?P5
(#11/#14/#21/#28/#30/#31), 6-language catalogs (#67). NEXT_STEPS.md:7 states 'khelsutra.guru is
LIVE and fully shipped'. UI_ALPHA_EXECUTION_PLAN.md:36 defines M1 only as the initial
'Vite+React+TS+Tailwind scaffold', which is long past.",
146         "suggestedFix": "Replace with a current status, e.g.: 'Status: alpha ? the
web/ SPA is built and multilingual (6 languages), with the per-rally selection MVP and first-run
setup wizard shipped; khelsutra.guru / app.khelsutra.guru are live (Cloudflare Pages +
cloudflared tunnel + Access). See NEXT_STEPS.md.'",
147         "severity": "high"
148     },
149     {
150         "claim": "the web app (alpha SPA) now, the hosted control plane at beta",
151         "docLocation": "README.md:4",
152         "verdict": "stale",
153         "evidence": "This tagline pairs with line 9's 'pre-alpha scaffold', which is
contradicted by the shipped, live SPA (see web/ evidence above and NEXT_STEPS.md:7). The 'alpha
SPA now' framing is directionally right but undercut by the stale 'pre-alpha scaffold' status it
sits beside.",
154         "suggestedFix": "Keep 'web app (alpha SPA) now' but align the rest of the
intro/status so it no longer describes the SPA as not-yet-landed.",
155         "severity": "medium"
156     },
157     {
158         "claim": "`control-plane/` ? (beta) Cloud Run FastAPI control plane",
159         "docLocation": "README.md:17",
160         "verdict": "stale",
161         "evidence": "No control-plane/ directory exists: `ls control-plane` -> 'No
such file or directory'; `git ls-files control-plane/` returns nothing. Top-level tracked dirs
are deploy/, docs/, locales/, site/, web/ only.",
162         "suggestedFix": "Mark it as a future/planned directory that does not yet
exist (e.g. move it out of the 'Structure' list into a 'Planned (beta)' note), so a reader
doesn't expect the path to be present.",
163         "severity": "medium"
164     },
165     {
166         "claim": "Engine ... sibling repo `badminton-highlight-indexer` (rename
pending).",

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167         "docLocation": "README.md:6",
168         "verdict": "correct",
169         "evidence": "Sibling repo still exists under that name: `ls
C:/Users/avidu/Projects/` shows 'badminton-highlight-indexer'; no rename has occurred, so
'rename pending' holds.",
170         "suggestedFix": "",
171         "severity": "low"
172     },
173     {
174         "claim": "Canonical planning docs live in the engine repo's docs/ for now:
UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md.",
175         "docLocation": "README.md:7-8",
176         "verdict": "correct",
177         "evidence": "All three exist in the engine repo (badminton-highlight-
indexer/docs/UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md present) and are NOT
tracked in khelsutra/docs (`git ls-files docs/` in khelsutra returns none of them; `ls
docs/UI_PROPOSAL.md` -> not found).",
178         "suggestedFix": "",
179         "severity": "low"
180     },
181     {
182         "claim": "`deploy/` ... alpha deploy kit ... DEPLOY_ALPHA.md ... +
cloudflared.config.example.yml",
183         "docLocation": "README.md:15",
184         "verdict": "correct",
185         "evidence": "git ls-files deploy/ shows deploy/DEPLOY_ALPHA.md and
deploy/cloudflared.config.example.yml (plus deploy/droplet/*). DEPLOY_ALPHA.md:1-5 matches the
described 'shareable page, no SSH; SPA on Pages + engine on api.* tunnel + CF Access' content.",
186         "suggestedFix": "",
187         "severity": "low"
188     },
189     {
190         "claim": "`site/` ? public marketing landing page (static; served at
khelsutra.guru)",
191         "docLocation": "README.md:13",
192         "verdict": "correct",
193         "evidence": "site/ exists with site/public/index.html, server.js, etc.;
DEPLOY_ALPHA.md and NEXT_STEPS.md confirm the marketing site is served at khelsutra.guru (apex)
while the SPA is at app.khelsutra.guru.",
194         "suggestedFix": "",
195         "severity": "low"
196     },
197     {
198         "claim": "`web/` ? React SPA (alpha; Vite + React + TS + Tailwind;
Cloudflare Pages build root)",
199         "docLocation": "README.md:14",
200         "verdict": "correct",
201         "evidence": "web/ is a Vite+React+TS+Tailwind SPA (web/README.md:3 'Vite +
React + TS + Tailwind v4'); DEPLOY_ALPHA.md:14 confirms 'app.khelsutra.guru -> Cloudflare Pages
(the web/ SPA)'. The tech-stack/build-root facts are accurate; only the inline 'alpha' maturity
ties into the stale status above.",
202         "suggestedFix": "",
203         "severity": "low"
204     },
205     {
206         "claim": "License: All rights reserved (private; no OSS license).",
207         "docLocation": "README.md:19-21",
208         "verdict": "correct",
209         "evidence": "No LICENSE file is tracked (not present in `git ls-files`);
consistent with a private 'all rights reserved' repo.",
210         "suggestedFix": "",
211         "severity": "low"
212     }
213 ]
214 },

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215     {
216         "doc": "deploy/DEPLOY_ALPHA.md",
217         "overallStatus": "minor-staleness",
218         "summary": "Verified correct except a stale header",
219         "findings": [
220             {
221                 "claim": "header stale deploy live",
222                 "docLocation": "deploy/DEPLOY_ALPHA.md:3",
223                 "verdict": "stale",
224                 "evidence": "droplet/README.md:100 live-verified 2026-06-22",
225                 "suggestedFix": "set header to LIVE",
226                 "severity": "medium"
227             },
228             {
229                 "claim": "host list too narrow",
230                 "docLocation": "deploy/DEPLOY_ALPHA.md:66",
231                 "verdict": "stale",
232                 "evidence": "engine.env.example adds loopback",
233                 "suggestedFix": "add localhost",
234                 "severity": "low"
235             }
236         ]
237     },
238     {
239         "doc": "deploy/droplet/README.md",
240         "overallStatus": "fresh",
241         "summary": "This command-dense droplet quickstart is accurate and matches the
live repo: every cited kit file exists with the described contents, and the engine-behavior
claims (fail-fast boot posture, RALLY_* exposure vars, the Gemini-default-on-CPU gotcha, the
gs:// jail rejection, the --server entryptpoint) all verify against the engine repo and
DEPLOY_ALPHA.md. The only items not locally reproducible are the 2026-06-22 droplet live-run
figures (e.g. \"1313 passed\"), which are an off-box assertion ? they are internally plausible
(engine has ~1269 test functions before parametrization) but cannot be confirmed from these
repos.",
242         "findings": [
243             {
244                 "claim": "Four kit files exist: config.json.example, engine.env.example,
khelsutra-engine.service (and ../cloudflared.config.example.yml), each going to the documented
destination with the documented purpose.",
245                 "docLocation": "deploy/droplet/README.md:9-14",
246                 "verdict": "correct",
247                 "evidence": "git ls-files deploy/ lists all four:
deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service} and
deploy/cloudflared.config.example.yml; contents read and match (config.json.example has the
security block + cf_team/aud; engine.env.example has the two RALLY_* exposure vars +
RALLY_APP_HOME; the service binds 127.0.0.1:8000; cloudflared.config routes api. ->
127.0.0.1:8000).",
248                 "suggestedFix": "",
249                 "severity": "low"
250             },
251             {
252                 "claim": "The boot fails fast (exit 1, '[STARTUP] refusing to start') if
edge_auth_enabled=true but the jail or cf_team_domain/cf_access_aud is unset.",
253                 "docLocation": "deploy/droplet/README.md:53-54",
254                 "verdict": "correct",
255                 "evidence": "backend/main.py:857 logs '[STARTUP] refusing to start';
backend/config/startup.py:109 EXPOSED_NO_VIDEO_ROOT and :123 EXPOSED_EDGE_AUTH_INCOMPLETE are
both LEVEL_ERROR (startup.py:41,272-276), while EXPOSED_AUTH_EXTRA_MISSING (PyJWT) is LEVEL_WARN
(:135). From engine PR #294 (commit 0c0e064).",
256                 "suggestedFix": "",
257                 "severity": "low"
258             },
259             {
260                 "claim": "indexing.default_segmenter MUST be gemini on a CPU box; the
shipped default yolo_hybrid needs torch/GPU and every job fails 'No module named torch' without

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it.",
261         "docLocation": "deploy/droplet/README.md:109-113",
262         "verdict": "correct",
263         "evidence": "config.json.example:9-11 sets default_segmenter=gemini. Engine
AppConfig default is yolo_hybrid (backend/config/models.py:128), so an all-defaults load on the
droplet (which hand-writes config.json rather than running the --app LIGHT seeder) does land on
yolo_hybrid. The LIGHT 'motion' default (loader.py:227) only applies via
write_light_default_config (indexer --app first-run, PR #310), not this systemd path ? so the
gotcha is accurate for this deploy.",
264         "suggestedFix": "",
265         "severity": "low"
266     },
267     {
268         "claim": "RALLY_ALLOWED_HOSTS is the load-bearing fix; with edge_auth on,
the Host-header guard defaults loopback-only and 400s every tunneled request until set.",
269         "docLocation": "deploy/droplet/README.md:92, engine.env.example:14-17",
270         "verdict": "correct",
271         "evidence": "backend/config/startup.py:143-147 raises
EXPOSED_HOST_ALLOWLIST_LOOPBACK (LEVEL_WARN) when edge_auth is ON and RALLY_ALLOWED_HOSTS is
unset, with the exact 'will 400 forwarded requests' rationale; backend/main.py reads
RALLY_ALLOWED_HOSTS / RALLY_BIND_HOST / RALLY_CORS_ALLOW_ORIGINS. Matches DEPLOY_ALPHA.md §2.2
repro (400->200).",
272         "suggestedFix": "",
273         "severity": "low"
274     },
275     {
276         "claim": "Pasting a gs:// URI is rejected by the jail on the exposed box;
upload is the alpha path.",
277         "docLocation": "deploy/droplet/README.md:96-98",
278         "verdict": "correct",
279         "evidence": "tests/test_api_endpoints.py:608
test_process_hosted_mode_rejects_nonlocal_uri pins the gs:// -> 400 behavior under
allowed_video_root (referenced at :541 as the counterweight); the jail is required to expose
(config.json.example sets allowed_video_root).",
280         "suggestedFix": "",
281         "severity": "low"
282     },
283     {
284         "claim": "Pages build: Root dir web, Build npm run build, Output web/dist,
with VITE_API_BASE = https://api.khelsutra.guru/api.",
285         "docLocation": "deploy/droplet/README.md:73-76",
286         "verdict": "correct",
287         "evidence": "khelsutra web/ SPA exists (git ls-files web/...);
web/package.json build script = 'vite build'; web/src/api/client.ts:25 reads
import.meta.env.VITE_API_BASE (default '/api'), and client.test.ts:32 exercises the exact split
value https://api.khelsutra.guru/api. Vite default outDir 'dist' relative to root web/ =
web/dist.",
288         "suggestedFix": "",
289         "severity": "low"
290     },
291     {
292         "claim": "The shipped engine is run via the auth extra and python -m
backend.main --server, binding 127.0.0.1:8000.",
293         "docLocation": "deploy/droplet/README.md:33,47-49; khelsutra-
engine.service:26",
294         "verdict": "correct",
295         "evidence": "backend/main.py:867 defines the --server flag and :1107 calls
uvicorn.run(app, host=bind_host, port=port); --setup flag at :865; the service ExecStart uses
python -m backend.main --server.",
296         "suggestedFix": "",
297         "severity": "low"
298     },
299     {
300         "claim": "Live-verified on the droplet (2026-06-22): full suite 1313 passed
+ ruff/mypy clean on the box; boot-posture, Host-guard 400->200, path-jail, T4 upload->jail,

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edge-auth 401-gate exercised live; full DNS->Cloudflare->Access->tunnel->engine chain
confirmed.",
301         "docLocation": "deploy/droplet/README.md:100-106",
302         "verdict": "unverifiable",
303         "evidence": "This is an off-box live-run assertion that cannot be reproduced
from the khelsutra or engine repos here. Sanity check: the engine tests/ dir has ~1269 'def
test_' functions (parametrize expansions push the collected/passed count higher), so '1313
passed' is plausible and consistent, not contradicted. Marked unverifiable per the file:line-
evidence discipline; no sign of staleness.",
304         "suggestedFix": "",
305         "severity": "low"
306     },
307     {
308         "claim": "Cost: Cloudflare Pages + Tunnel + Access are free (Access <= 50
users); standing cost ~ $0 + Gemini per run.",
309         "docLocation": "deploy/droplet/README.md:16-18",
310         "verdict": "unverifiable",
311         "evidence": "External Cloudflare pricing claim; not checkable against the
repo. Consistent with DEPLOY_ALPHA.md:3 ('~$0/mo standing ... + Gemini per-run'). No internal
contradiction.",
312         "suggestedFix": "",
313         "severity": "low"
314     }
315 ]
316 },
317 {
318     "doc": "docs/ANNOTATE_AND_TRAIN.md",
319     "overallStatus": "significant-staleness",
320     "summary": "A DRAFT tracker whose marketing-page deliverables (P0-P4 plus the
real annotator screenshots P5 mentions) are all shipped and merged (#15/#22/#23/#25), yet the
doc was last touched 2026-06-20 (#23) and never updated to reflect that. The core
engine/annotator technical claims (schema bridge, ship-gate refusing a commercial ship, MIT
annotator v1.6.4, guardrails) all still verify as correct ? only their file:line citations have
drifted. Main staleness: status still DRAFT despite shipped work, P5 marked Todo though
annotator screenshots are committed, the doc is NOT actually indexed in docs/README.md (claimed
three times), and the page since grew 6-language localization the tracker never anticipated. It
is an archive candidate once honestly closed out.",
321     "findings": [
322     {
323         "claim": "Status: DRAFT ? owner-gated on scope (header, line 3); Last
updated: 2026-06-20.",
324         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:3",
325         "verdict": "stale",
326         "evidence": "git log --follow shows the marketing-page work merged: P0 doc
(#15, 6b62ebf), own-model page P1-P4 (#22, b1110e1), org link (#23, 87520b8). Per $7, P0-P4 are
all marked Done ? yet the header still reads DRAFT. Lifecycle (line 4) is DRAFT->IN
PROGRESS->DONE; with P0-P4 shipped this is at least IN PROGRESS.",
327         "suggestedFix": "Flip Status to IN PROGRESS (only owner-gated P5-P9 remain)
and bump Last updated; or, if the owner considers the marketing surface complete, mark DONE and
archive per the lifecycle note.",
328         "severity": "medium"
329     },
330     {
331         "claim": "P5 'Capture + commit real screenshots ... replace placeholders' ?
Status ? (Todo), Gated Owner.",
332         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:76",
333         "verdict": "stale",
334         "evidence": "PR #25 (1906c82) 'show the real annotator screenshots'
committed site/public/img/annotate-enable.png (133712 bytes) and annotate-tool.png (153133
bytes); own-model.html:111-112 references them as the real VLC-UI shots (no people/footage). The
annotator-UI half of P5 is done and merged, not Todo.",
335         "suggestedFix": "Mark P5 ? for the annotator-UI screenshots (cite #25); if
footage screenshots with consented people are still pending, split that into its own row so the
done part isn't hidden under a blanket Todo.",
336         "severity": "high"

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337         },
338         {
339             "claim": "Doc is 'index[ed] in docs/README.md' (line 5), §7 P0 'indexed in
docs/README.md' (? , #15), and DoD 'doc indexed in docs/README.md'.",
340             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:5,71,91",
341             "verdict": "stale",
342             "evidence": "grep -ni 'annotate|own.model|train.your|ANNOTATE_AND'
docs/README.md returns NOT FOUND; the docs/README.md 'Tracked work' list (lines 1-30) does not
mention ANNOTATE_AND_TRAIN.md or own-model. The claimed index entry does not exist.",
343             "suggestedFix": "Either add an ANNOTATE_AND_TRAIN.md entry to
docs/README.md's 'Tracked work' list, or correct P0/DoD to reflect that it is not yet indexed.",
344             "severity": "high"
345         },
346         {
347             "claim": "Engine ship-gate (gate_verdict, harness.py:98) currently refuses a
commercial ship regardless of F1: WASB provenance C3 unresolved + F1@tIoU not calibrated
(#172).",
348             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:12,31",
349             "verdict": "correct",
350             "evidence": "training/gen0/harness.py:177 returns {'ship': False, ...}; :175
'WASB checkpoint provenance C3 UNRESOLVED ? commercial ship blocked regardless of F1'; :174
'ship-gate F1@tIoU target NOT YET CALIBRATED ... see issue #172'. Fact correct; the cited line
(:98) has drifted ? gate_verdict is now at :142.",
351             "suggestedFix": "Update the line citations: gate_verdict is at
harness.py:142 (not :98).",
352             "severity": "low"
353         },
354         {
355             "claim": "Default synth path: synth_trajectory_from_labels uses
csv.DictReader reading row['start_time']/row['end_time'] (stub_runner.py:139) ? MATCHES the
annotator header exactly.",
356             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:46",
357             "verdict": "correct",
358             "evidence": "backend/pipeline/detectors/stub_runner.py:124 def
synth_trajectory_from_labels; :137 for row in _csv.DictReader(f); :139
rallies.append((float(row['start_time']), float(row['end_time']))). Annotator header
rally_number,start_time,end_time,... confirmed at rally-annotator vlc/rally_annotator.lua:128
and docs/CSV_FORMAT.md:6. Claim accurate.",
359             "suggestedFix": "",
360             "severity": "low"
361         },
362         {
363             "claim": "Canonical GT reader load_gt_intervals accepts start/end OR
start_time/end_time (gt_loader.py:14-90).",
364             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:47",
365             "verdict": "correct",
366             "evidence": "backend/eval/gt_loader.py:30 _START_NAMES = ('start',
'start_time'); :44 def load_gt_intervals; :70 headerless positional col0,col1 path. Both aliases
accepted as claimed.",
367             "suggestedFix": "",
368             "severity": "low"
369         },
370         {
371             "claim": "Bundle attests no_paid_llm_training_labels:True (harness.py:192)
and label source is human (harness.py:178).",
372             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:33,60",
373             "verdict": "correct",
374             "evidence": "training/gen0/harness.py:236 'no_paid_llm_training_labels':
True; :222 'label_source': 'human (rally-annotator golden CSVs)'. Facts correct; cited lines
(:192, :178) have drifted to :236 and :222.",
375             "suggestedFix": "Update line citations: no_paid_llm_training_labels at
harness.py:236, label_source at :222.",
376             "severity": "low"
377         },
378         {

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379         "claim": "cmd_train (__main__.py:231) pulls labels/, joins video<->label by
file MD5 + stem-match labels/<name>.rallies.csv <-> videos/<name>.<ext> (__main__.py:271-280).",
380         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:31,49",
381         "verdict": "correct",
382         "evidence": "backend/worker/__main__.py:332 def cmd_train; :359 lists
{corpus}/labels/; :383-384 fname.replace('.rallies.csv','') stem join. Logic present and
correct; cmd_train line drifted from :231 to :332.",
383         "suggestedFix": "Update cmd_train citation to __main__.py:332.",
384         "severity": "low"
385     },
386     {
387         "claim": "Reuse map: add a /own-model alias in site/server.js:49-51; Worker
env.ASSETS.fetch needs no change (site/src/index.js:84).",
388         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52,73",
389         "verdict": "correct",
390         "evidence": "site/server.js:99 'pathname === \"/own-model\" ? \"/own-
model.html\"' and :41 LOCALIZED_PAGES includes own-model.html (alias shifted off the cited
:49-51). site/src/index.js:84 'return env.ASSETS.fetch(request)' with no /own-model branch ?
confirms the Worker needs no change.",
391         "suggestedFix": "Update the server.js alias citation (now ~:99, not
:49-51).",
392         "severity": "low"
393     },
394     {
395         "claim": "Reuse precedent: screenshots under site/public/img/ (precedent
site/og.png + make_og.py).",
396         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52",
397         "verdict": "correct",
398         "evidence": "git ls-files: site/make_og.py and site/public/og.png both
exist; site/public/img/ holds annotate-enable.png, annotate-tool.png, studio-*.png. The og asset
path is site/public/og.png (the doc wrote 'site/og.png' loosely), make_og.py is at
site/make_og.py.",
399         "suggestedFix": "Minor: the OG image is at site/public/og.png (not
site/og.png).",
400         "severity": "low"
401     },
402     {
403         "claim": "SPA today: web/src/App.tsx is one static screen, no router
(web/package.json); a correction screen needs react-router-dom.",
404         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,64",
405         "verdict": "stale",
406         "evidence": "grep on web/package.json: no react-router dep (claim 'no
router' still TRUE). But App.tsx is no longer 'one static screen' ? it now imports and
conditionally renders SetupWizard (web/src/App.tsx:11,65,242, P2c first-run wizard from PR #88).
The 'one static screen' framing predates the wizard.",
407         "suggestedFix": "Reword to 'no router; App.tsx renders a first-run
SetupWizard + the ingest panel' so the SPA surface description matches the post-#88 code.",
408         "severity": "low"
409     },
410     {
411         "claim": "SPA correction route depends on a new, unbuilt engine endpoint
PATCH /api/segments/{id} (dead columns
candidate_segments.human_label/notes/confirmed_segment_id); P8 ? Gated.",
412         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,79",
413         "verdict": "correct",
414         "evidence": "grep across backend/ for 'api/segments'/'PATCH' on
candidate_segments finds only the candidate_segments table + query_candidate_segments readers
(database.py:217, eval/*.py); no PATCH /api/segments route exists. P8 is still genuinely unbuilt
as stated.",
415         "suggestedFix": "",
416         "severity": "low"
417     },
418     {
419         "claim": "Relation note: peer of LOCAL_APPLIANCE_ARCHITECTURE.md (the latter
lands with PR #13).",

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420         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:6,84",
421         "verdict": "stale",
422         "evidence": "git ls-files confirms docs/LOCAL_APPLIANCE_ARCHITECTURE.md
already exists in the repo; it has long since landed, so the forward-looking 'lands with PR #13'
/ 'cross-link once PR #13 merges' phrasing is outdated.",
423         "suggestedFix": "Drop the 'lands with PR #13' future-tense framing ? the doc
exists now; add the cross-link.",
424         "severity": "low"
425     },
426     {
427         "claim": "rally-annotator is a single-file VLC Lua extension, VERSION 1.6.4,
MIT-licensed, public at github.com/Khelsutra/rally-annotator.",
428         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:29",
429         "verdict": "correct",
430         "evidence": "rally-annotator/vlc/rally_annotator.lua:1 '(v1.6.4)'; LICENSE
first line 'MIT License'; own-model.html:109/191 links to https://github.com/Khelsutra/rally-
annotator (org-link migration was PR #23, captured in the doc's own history).",
431         "suggestedFix": "",
432         "severity": "low"
433     },
434     {
435         "claim": "$7 tracker: P1/P2/P3/P4 all ? via '(this PR)'.",
436         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:72-75",
437         "verdict": "correct",
438         "evidence": "own-model.html exists (P1); /own-model alias in server.js:99
(P2); nav cross-link index.html:255 + footer :620 (P3); verbatim annotator content + GitHub link
in own-model.html:108-191 (P4). All shipped ? git history shows these in PR #22 (b1110e1,
'annotate-train P1-P4'), so '(this PR)' should now read '#22'.",
439         "suggestedFix": "Replace the '(this PR)' placeholders in the PR column with
#22 (and #23 for the org-link follow-up) so the tracker cites real merged PRs.",
440         "severity": "medium"
441     }
442 ]
443 },
444 {
445     "doc": "docs/DRY_RUN_CUJ.md",
446     "overallStatus": "minor-staleness",
447     "summary": "The doc's substantive engineering claims are accurate and well-
grounded: the SPA back-half + B-P2 front-half ingest screen exist
(web/src/components/IngestPanel.tsx, web/src/hooks/useIngestJob.ts, #56), the engine bucket-
ingest routing is verified in code, and the three cited engine PRs (#280/#281/#289) plus all
four cited engine tests exist. The staleness is in metadata/framing: the header still says
Status: DRAFT though docs/README.md correctly calls it DELIVERED with DoD $6 boxes 1-3 done, and
the doc repeatedly describes the multilingual workflow as \"en/kn/hi\" (3 langs) when the SPA
now supports 6 (es/da/id added in #87). It is a completed tracker and an archive candidate once
the parked droplet box is closed.",
448     "findings": [
449         {
450             "claim": "Status: DRAFT ? owner-gated . ... Goal for this session.",
451             "docLocation": "docs/DRY_RUN_CUJ.md:3",
452             "verdict": "stale",
453             "evidence": "docs/README.md:4 marks this doc '? DELIVERED (2026-06-21)'; DoD
$6 boxes 1-3 are all checked [x] (DRY_RUN_CUJ.md:47-49) and the work shipped (#53/#55/#56/#57).
A DRAFT/'goal for this session' header no longer matches the completed, delivered state.",
454             "suggestedFix": "Flip the header to Status: DONE / DELIVERED (2026-06-21)
and drop 'Goal for this session'; note the only open box (droplet, $6 box 4) is owner-parked.
Consider archiving once that box closes.",
455             "severity": "medium"
456         },
457         {
458             "claim": "The SPA is 'Fully en/kn/hi (the switcher / ?lang=)' and the
multilingual workflow is en/kn/hi.",
459             "docLocation": "docs/DRY_RUN_CUJ.md:11",
460             "verdict": "stale",
461             "evidence": "web/src/i18n/index.ts:25 ? `export const SUPPORTED_LOCALES =

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[\"en\\\", \"kn\\\", \"hi\\\", \"es\\\", \"da\\\", \"id\\\"]`; es/da/id were added to the SPA in PR #87
(cf59fbf 'add es/da/id to the marketing site (parity with the SPA's 6 languages)'). The SPA now
supports 6 languages, not 3.",
462     "suggestedFix": "Update 'en/kn/hi' to 'en/kn/hi/es/da/id (6 languages)'
where it describes the current switcher; the Hindi/Kannada run results in §7 stay accurate as
historical record.",
463     "severity": "low"
464 },
465 {
466     "claim": "This SPA front-half (ingest/process UI) is the *only* net-new work
for Path B (= B-P2); the SPA has no ingest/process UI today.",
467     "docLocation": "docs/DRY_RUN_CUJ.md:14",
468     "verdict": "correct",
469     "evidence": "B-P2 was net-new work at draft time and has since shipped:
web/src/components/IngestPanel.tsx + web/src/hooks/useIngestJob.ts (first landed in PR #56
'feat(web): B-P2 ? in-UI bucket ingest', 4525c5d, 2026-06-21). useIngestJob.ts drives POST
/api/process (processVideo) ? poll GET /api/jobs/{id} (getJob) ? hand video_id to App, exactly
as scoped.",
470     "suggestedFix": "",
471     "severity": "low"
472 },
473 {
474     "claim": "POST /api/process accepts gs://gcs://s3://local:// URIs and skips
the local-existence check for non-local backends; resolve via storage.resolve_input_ref;
gs://?gcs normalization.",
475     "docLocation": "docs/DRY_RUN_CUJ.md:15,68-72",
476     "verdict": "correct",
477     "evidence": "Engine backend/main.py:581 `input_ref =
storage.resolve_input_ref(req.video_path, ...)` and main.py:590 `if input_ref.backend ==
'local' and not os.path.exists(input_ref.key):` (the non-local skip), with the fetch at
main.py:458 `storage.acquire_local_input(...)`; backend/storage/ref.py:60 `if scheme == 'gs':
scheme = 'gcs'`. Substance matches; the exact line numbers 159/274/283 cited in the doc have
drifted as the engine shipped further PRs.",
478     "suggestedFix": "Optionally refresh the cited line numbers (now
~main.py:581/590/458, ref.py:60) since the engine surface moved; the routing logic itself is
intact.",
479     "severity": "low"
480 },
481 {
482     "claim": "Tested by
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches,
::test_process_hosted_mode_rejects_nonlocal_uri, ::test_process_unknown_scheme_returns_400,
::test_cli_nonlocal_input_fetches_and_keeps_uri.",
483     "docLocation": "docs/DRY_RUN_CUJ.md:15,76",
484     "verdict": "correct",
485     "evidence": "Engine repo grep: tests/test_api_endpoints.py:508
test_process_nonlocal_uri_skips_exists_and_fetches; :577
test_process_unknown_scheme_returns_400; :608 test_process_hosted_mode_rejects_nonlocal_uri;
:627 test_cli_nonlocal_input_fetches_and_keeps_uri. All four exist with the asserted names.",
486     "suggestedFix": "",
487     "severity": "low"
488 },
489 {
490     "claim": "B-P1 (gs://?/api/process?API-DB video_id) already shipped in
engine M2 / #280; the job-drain hang fixed in #281; the compile-from-bucket 500 fixed in #289.",
491     "docLocation": "docs/DRY_RUN_CUJ.md:15,63,68,88",
492     "verdict": "correct",
493     "evidence": "Engine git log: e111add 'feat(serving): M2 ? URI-aware input
... (#280)'; 41d45fd 'fix(server): drain async jobs under `python -m backend.main` (double-
module hang) (#281)'; e68b583 'fix(api): resolve a bucket-ingested video's source to a local
file before stitching (#289)'. All three PRs exist with descriptions matching the doc.",
494     "suggestedFix": "",
495     "severity": "low"
496 },
497 {

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498         "claim": "DoD §6 box 3: in-UI Path-B bucket?reel CUJ DONE (live e2e
2026-06-21) ? 12 rallies loaded ? reel downloaded; box 4 (CPU droplet) still open/un-
provisioned.",
499         "docLocation": "docs/DRY_RUN_CUJ.md:49-50",
500         "verdict": "correct",
501         "evidence": "Box 3 is corroborated by PR #57 'docs(dry-run): Path-B live e2e
DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3 ?)' and the merged B-P2 SPA (#56).
docs/README.md:6-7 confirms box 4 (droplet, option 3) remains 'parked by the owner (GPU
subscription)'. The run-result specifics (12 rallies, reel sizes, GCS screencast paths) are
runtime artifacts not locally reproducible.",
502         "suggestedFix": "",
503         "severity": "low"
504     },
505     {
506         "claim": "Run-result specifics: 12 rallies / video_id 05e0e577 /
50.9MB+76.5s reels / screencasts at gs://khelsutra-rally-corpus/promo/*.mp4 and the staged
bucket video gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4.",
507         "docLocation": "docs/DRY_RUN_CUJ.md:54-60,82-85",
508         "verdict": "unverifiable",
509         "evidence": "These are GCS bucket objects and a one-off local run's outputs;
they cannot be checked from the repo. No counter-evidence found ? the surrounding code/PR claims
that frame them are all verified.",
510         "suggestedFix": "",
511         "severity": "low"
512     }
513 ]
514 },
515 {
516     "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
517     "overallStatus": "significant-staleness",
518     "summary": "A still-active DRAFT tracker (last updated 2026-06-21) whose §9
progress and several \"Absent (net-new)\" engine-fact anchors have drifted: the engine has since
shipped /api/capabilities, exposure-safety boot posture + an edge-auth layer, and merged
native_wasb_runner to master, while the alpha deploy went LIVE on 2026-06-22
(app.khelsutra.guru). The khelsutra-side claims (PR refs #11/#56/#64/#66, IngestPanel,
VITE_API_BASE) check out; the staleness is concentrated in engine status and the deploy/DoD
progress. Not yet an archive candidate (open owner gates remain) but needs a refresh pass.",
519     "findings": [
520     {
521         "claim": "§3 lists `GET /api/capabilities` under 'Absent (net-new): ...
`[verified ? grep returns none]`; §4.6 Reuse map lists it under 'Net-new'; §9 T7 (capabilities
+ health + banner) is ? Todo.",
522         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:49,113,164",
523         "verdict": "stale",
524         "evidence": "engine origin/master backend/main.py:340
`@app.get(\"/api/capabilities\")` (and main.py:307 `/api/health`, main.py:334 `/api/version`)
all exist on master. git grep '/api/capabilities' origin/master returns a hit.",
525         "suggestedFix": "Move `/api/capabilities` out of 'Absent (net-new)' and out
of §4.6 'Net-new'; mark the engine half of T7 (capabilities+health) as shipped (?/?), leaving
only the SPA banner pending.",
526         "severity": "high"
527     },
528     {
529         "claim": "TL;DR + Risk table: 'the engine has zero app-layer auth and CORS
`*` `[verified backend/main.py:42]` ? structural safety only because the engine binds
127.0.0.1.",
530         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:14,30,103,128",
531         "verdict": "stale",
532         "evidence": "engine origin/master backend/main.py:82 uses
`allow_origins=cors_origins_from_env()` (default ['*'] but env-lockable) with
`allow_credentials=False`; main.py:109 adds TrustedHostMiddleware (DNS-rebinding/Host
allowlist); backend/api/auth.py + config/startup.py enforce_startup_checks + M9 `Cf-Access-
Authenticated-User-Email` `owner_id` now exist on master. CORS is no longer a hardcoded bare
`*`, and an app-layer edge-auth integration now exists.",
533         "suggestedFix": "Reword to: CORS default is `*` but now env-configurable

```

```

(allow_credentials off), and the engine now ships a Host-header guard + an edge-auth integration
(Cf-Access trust) + exposure-safety boot posture ? so 'zero app-layer auth' is no longer
literally true. Re-anchor the line (main.py:42 no longer points at CORS).",
534         "severity": "high"
535     },
536     {
537         "claim": "$3 / $4.3: 'Native WASB is in-flight on branch `feat/native-wasb-
runner`: `native_wasb_runner.py` exists `[verified]`'.",
538         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:45,96",
539         "verdict": "stale",
540         "evidence": "engine origin/master now contains
backend/pipeline/detectors/native_wasb_runner.py and tests/test_native_wasb_runner.py (git ls-
tree origin/master). The runner has merged to master; the `feat/native-wasb-runner` branch still
exists but the file is no longer branch-only. (trajectory_hybrid._build_native_runner still
raises NotImplementedError at trajectory_hybrid.py:61 ? that sub-claim is still correct.)",
541         "suggestedFix": "Change 'in-flight on branch feat/native-wasb-runner' to
'native_wasb_runner.py is on master; trajectory_hybrid still refuses, fusion builds it; owner
byte-parity gate still pending.'",
542         "severity": "medium"
543     },
544     {
545         "claim": "$9 alpha-deploy block: 'the actual stand-up (tunnel + Pages + CF
Access + key rotation) is the **owner** step'; $11 DoD box 1 (CUJ-1 e2e on droplet behind gate)
and box 2 (raw :8000 refused, tested) unchecked.",
546         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:175,182,183",
547         "verdict": "stale",
548         "evidence": "deploy/droplet/README.md:100-106 '? Live-verified on the
droplet (2026-06-22)': engine booted on 168.144.126.176; suite 1313 passed; boot-posture fail-
fast, Host-guard 400?200, path-jail, T4 upload?jail, and edge-auth 401-gate all exercised live;
full DNS?Cloudflare?Access?tunnel?engine chain confirmed; app.khelsutra.guru up. Only Gemini-key
rotation + a browser OTP login remain.",
549         "suggestedFix": "Update $9 to note the alpha stand-up is LIVE (2026-06-22,
app.khelsutra.guru, posture test exercised) with only key rotation + browser-OTP login
outstanding; tick/qualify the corresponding DoD boxes.",
550         "severity": "high"
551     },
552     {
553         "claim": "Status header: 'Status: DRAFT ... Last updated: 2026-06-21' with
$9 the source of truth.",
554         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:3",
555         "verdict": "stale",
556         "evidence": "khelsutra git log shows merges through #94 (master HEAD
50b2bd7) and the alpha went live 2026-06-22, both after the doc's 2026-06-21 'last updated'.
Multiple $9 rows and engine facts have moved since (T7 engine half, alpha deploy, native_wasb on
master).",
557         "suggestedFix": "Bump 'Last updated' and consider promoting DRAFT?IN
PROGRESS; refresh $9 rows to current reality.",
558         "severity": "medium"
559     },
560     {
561         "claim": "D4 'Embed RallyPicker ... PR [#11]' and $9 T4 '[#66]', T12 'SPA
#56', alpha-deploy '[#64]' / 'VITE_API_BASE'.",
562         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:33,161,169,175",
563         "verdict": "correct",
564         "evidence": "khelsutra git log: cb74ce5 '...RallyPicker (#11)'; ca619fb 'T4
? in-UI file upload (#66)'; 4525c5d 'B-P2 ? in-UI bucket ingest (#56)'; 8bbc557 'VITE_API_BASE
(#64)'. web/src/components/IngestPanel.tsx handles both chunked upload (T4) and a gs:// URI
(T12); web/src/api/client.ts:25 reads VITE_API_BASE.",
565         "suggestedFix": "",
566         "severity": "low"
567     },
568     {
569         "claim": "T12 (bucket-URI ingest) marked ?: 'engine accepts URIs (M2/#280) +
SPA ingest screen shipped (B-P2, #56); in-UI gs:///reel CUJ proven live e2e (DRY_RUN_CUJ.md
$9)'.",

```

```

570         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:169",
571         "verdict": "correct",
572         "evidence": "docs/DRY_RUN_CUJ.md:80-92 §9 records the live in-UI gs://?reel
CUJ (12 rallies ? reel downloaded) and the compile-from-bucket 500 fix (engine #289);
IngestPanel.tsx accepts a gs:// URI. khelsutra #56 (B-P2) is merged.",
573         "suggestedFix": "",
574         "severity": "low"
575     },
576     {
577         "claim": "T11 retention marked ?: sweep tool shipped (engine #299), auto-
scheduling tracked (engine #301).",
578         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:168,130",
579         "verdict": "unverifiable",
580         "evidence": "These reference the engine repo's GitHub PR #299 / issue #301
(Khelsutra/badminton-highlight-indexer). Not falsifiable from the local khelsutra checkout; the
local engine checkout is on a feature branch and I did not confirm the specific sweep tool by PR
number. The khelsutra-side row metadata is internally consistent (#69 flipped T11 to ? per git
log bea8a2a).",
581         "suggestedFix": "",
582         "severity": "low"
583     },
584     {
585         "claim": "§3 Droplet reality: co-hosts the marketing site at
`/srv/khelsutra-site`, node `:8787`, engine at `~/rally`; MinIO 9000/9001 + GCS emulator 9023.",
586         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48",
587         "verdict": "unverifiable",
588         "evidence": "These are live-droplet filesystem/port facts (168.144.126.176)
not checkable from the local repo. deploy/DEPLOY_ALPHA.md:31 and deploy/droplet/README.md:4
corroborate the IP/CPU-no-GPU/blr1 framing, but the exact paths/ports (/srv/khelsutra-site,
:8787, ~/rally) are not in-repo verifiable.",
589         "suggestedFix": "",
590         "severity": "low"
591     },

```

4. user (2026-06-22T09:46:28.339Z)

```

1      # Deploy KhelSutra Studio for alpha ? a shareable page, no SSH
2
3      > **Status:** `READY` (owner executes the cloud steps) · **Topology:** split ? SPA on
Cloudflare Pages + engine on an `api.*` cloudflared tunnel, Cloudflare Access over both (the
engine's `HOSTING_PLAN.md` §2 alpha plan). · **Cost:** ~$0/mo standing (Pages + Tunnel + Access
are free ?50 users) + Gemini per-run (~$0.05/match).
4
5      This turns the local-only flow (run engine + SPA on your box, drive it yourself) into a
**page you can share with alpha testers** ? they open `app.khelsutra.guru`, sign in (email
allowlist), point at a video, and get a reel. No SSH, no home IP exposed, no inbound ports.
6
7      ## 0. The shape
8
9      ```
10     tester browser
11     ? https
12     ?
13     Cloudflare edge (free): DNS · TLS · Access (email-OTP allowlist) in front of BOTH
hostnames
14     ??? app.khelsutra.guru ? Cloudflare Pages           (the web/ SPA; built with
VITE_API_BASE)
15     ??? api.khelsutra.guru ? cloudflared Tunnel (outbound-only) ?? 127.0.0.1:8000 on
YOUR box
16                                     ? FastAPI engine
17                                     ? (Gemini
segmenter; GPU optional)
18                                     ? async jobs,
SQLite, output/ on disk
19     ```

```



```

20
21     - **No inbound ports.** `cloudflared` dials out; the engine stays bound to
`127.0.0.1:8000`. The box's home/public IP is never exposed.
22     - **Auth at the edge.** Cloudflare Access gates both hostnames; the engine *also*
verifies the `Cf-Access` JWT in-app (defence in depth ? `security.edge_auth_enabled`), so a
direct hit to the tunnel can't spoof identity.
23     - **Cross-origin, on purpose.** `app.` (SPA) and `api.` (engine) are different origins ?
the SPA is built with `VITE_API_BASE=https://api.khelsutra.guru/api` (PR #64) and the engine
locks CORS to the SPA origin via `RALLY_CORS_ALLOW_ORIGINS` (no engine code change ? it's
already env-driven).
24
25     > **Alternative (single-origin):** if you'd rather run **one box** that serves the SPA
*and* the API behind one origin (Caddy reverse-proxy, no Pages, no cross-origin), see
`LOCAL_APPLIANCE_ARCHITECTURE.md` D1. This runbook ships the **split** topology you chose
(stable Pages-hosted page + a thin API tunnel).
26
27     ---
28
29     ## 1. Prerequisites
30
31     - A **box** to run the engine: the CPU droplet `168.144.126.176` (Gemini-only ? no GPU;
fine for alpha) **or** your Windows+WSL GPU box (real WASB). `ffmpeg` + Python 3.8+ installed;
the `badminton-highlight-indexer` engine checked out.
32     - A **Cloudflare account** with the `khelsutra.guru` zone (you already run the marketing
site there).
33     - `cloudflared` installed on the box (`https://developers.cloudflare.com/cloudflare-
one/connections/connect-networks/downloads/`).
34     - A **fresh** `GEMINI_API_KEY` (rotate the chat-shared one ? it uploads downsized chunks
to Google; this is a paid + cloud action).
35
36     ---
37
38     ## 2. Engine (the `api.` side)
39
40     ### 2.1 Install (with the auth extra)
41     ```bash
42     cd badminton-highlight-indexer
43     pip install -e .[auth]           # PyJWT[crypto] ? REQUIRED when edge auth is on, else
every authenticated request is rejected 401
44     ```
45
46     ### 2.2 The safe-exposed config
47     Create / edit `config.json` on the box. **Where it must live:** the engine reads
`$RALLY_APP_HOME/config.json` if `RALLY_APP_HOME` is set, else `./config.json` relative to *the
directory you launch the server from ? **not** necessarily the repo root. If the engine reads a
different (or absent) file it silently falls back to all-defaults, which means **edge auth is
OFF**. To make writer and reader agree by construction, write it with `python -m backend.main
--setup` (it targets the exact path the server reads) and edit *that* file:
48     ```jsonc
49     {
50         "security": {
51             "edge_auth_enabled": true,                // turn the CF-Access JWT verify ON
52             "allowed_video_root": "/srv/khelsutra/incoming", // the path-jail ? REQUIRED when
exposed
53             "upload_dir": "/srv/khelsutra/incoming/uploads", // MUST sit UNDER
allowed_video_root (see §7)
54             "cf_team_domain": "https://<your-team>.cloudflareaccess.com",
55             "cf_access_aud": "<the CF Access application AUD>" // filled in §4
56         },
57         "indexing": {
58             "default_segmenter": "gemini" // ? CPU box MUST set this ? the shipped default
`yolo_hybrid` needs torch/GPU, and the SPA sends no segmenter, so every job fails "No module
named 'torch'" without it
59         }
60     }

```

```

61     ``
62     And export the env (a `.env` or your shell):
63     ```bash
64     export GEMINI_API_KEY="<your-rotated-key>"
65     export RALLY_CORS_ALLOW_ORIGINS="https://app.khelsutra.guru"    # lock CORS to the SPA
origin
66     export RALLY_ALLOWED_HOSTS="api.khelsutra.guru"                # REQUIRED ? see ? below;
without it EVERY tunneled request 400s
67     export RALLY_BIND_HOST="127.0.0.1"                            # keep the loopback bind
(cloudflared dials 127.0.0.1:8000)
68     ``
69
70     > **? Two env vars you MUST set, or the tunnel is dead-on-arrival.** With
`edge_auth_enabled: true`, the engine (a) flips its bind to `0.0.0.0` and (b) keeps the DNS-
rebinding **Host-header guard** at its loopback-only default ? so it boots "healthy" but then
**400's every request `cloudflared` forwards with `Host: api.khelsutra.guru`** (including the
$6 `/api/health` check). The boot only *warns* about this (`EXPOSED_HOST_ALLOWLIST_LOOPBACK`),
it does not stop. Set **`RALLY_ALLOWED_HOSTS`="api.khelsutra.guru"``** (the load-bearing fix ? the
Host allowlist is independent of the bind) and **`RALLY_BIND_HOST`="127.0.0.1"``** (so the bind
stays loopback as $0/$6 claim; `cloudflared` dials loopback anyway). *Verified against engine
`origin/master`: with the allowlist unset, `GET /api/health` with `Host: api.khelsutra.guru` ?
`400`; with it set ? `200`.*
71
72     > **The engine refuses to start unsafe.** The boot posture (engine PR #294) **fails
fast** if `edge_auth_enabled` is on but the `allowed_video_root` jail or
`cf_team_domain`/`cf_access_aud` is missing, and **warns** if `PyJWT` isn't installed. So a
half-configured exposure can't silently come up ? fix what it prints and restart.
73
74     ### 2.3 Run
75     ```bash
76     python -m backend.main --server    # binds 127.0.0.1:8000 ? kept loopback by
RALLY_BIND_HOST ($2.2); with edge_auth ON and that var unset it would bind 0.0.0.0 instead
77     ``
78     Keep it alive with a process manager (`systemd` on the droplet, mirroring
`site/scripts/provision-vm.sh`'s unit; or `pm2`/`nssm` on Windows).
79
80     ---
81
82     ## 3. The SPA on Cloudflare Pages (the `app.` side)
83
84     The SPA ships as a **static Pages site**, git-connected like the marketing site (auto-
deploy on merge to `master`).
85
86     1. **Cloudflare dashboard ? Workers & Pages ? Create ? Pages ? Connect to Git** ? pick
the `khelsutra` repo.
87     2. **Build settings:**
88         - **Root directory:** `web`
89         - **Build command:** `npm run build`
90         - **Build output directory:** `web/dist`
91         - **Environment variable:** `VITE_API_BASE = https://api.khelsutra.guru/api` ? this
is what points the SPA at the engine tunnel (PR #64).
92     3. **Custom domain:** add `app.khelsutra.guru` to the Pages project.
93
94     > Every push to `master` now rebuilds + redeploys the SPA. To preview a change first,
Pages builds a per-PR preview URL.
95
96     ---
97
98     ## 4. Cloudflare Access (gate BOTH hostnames)
99
100    1. **Zero Trust ? Access ? Applications ? Add ? Self-hosted.**
101    2. **Application domain:** add **both** `app.khelsutra.guru` **and**
`api.khelsutra.guru` to the *same* application (one app ? one shared session cookie, so a tester
signs in once and both the SPA and its API calls are authorized).
102    3. **Policy:** Action *Allow*, rule *Emails* ? your alpha testers' addresses (free ? 50

```

```

users ? fits 10?25 testers). Use one-time-PIN (email OTP) so testers need no Cloudflare account.
103 4. **Copy the application `AUD` tag** ? paste into the engine's `config.json`
`security.cf_access_aud` (§2.2), and set `cf_team_domain` to `https://<your-
team>.cloudflareaccess.com`. Restart the engine.
104 5. **CORS preflight ?:** the browser sends an unauthenticated `OPTIONS` preflight to
`api.` for the cross-origin POSTs. In the Access application's **Settings ? enable ?Bypass
options requests to CORS preflight?** (or add an explicit OPTIONS bypass), else the preflight is
redirected to the login page and every API call fails. The engine's CORS middleware
(`allow_credentials=false`, no cookies) then answers the preflight.
105
106 ---
107
108 ## 5. The tunnel (engine ? Cloudflare)
109
110 ```bash
111 cloudflared tunnel login # browser auth to your zone
112 cloudflared tunnel create khelsutra-studio # note the <TUNNEL_UUID>
113 # copy deploy/cloudflared.config.example.yml ? /etc/cloudflared/config.yml, fill
<TUNNEL_UUID>
114 cloudflared tunnel route dns khelsutra-studio api.khelsutra.guru
115 cloudflared tunnel run khelsutra-studio # or install as a service:
`cloudflared service install`
116 ```
117 See `deploy/cloudflared.config.example.yml` for the ingress (only `api.khelsutra.guru ?
http://127.0.0.1:8000`).
118
119 ---
120
121 ## 6. Verify (the posture test + the CUJ)
122
123 **Exposure posture (must hold ? this is what makes it safe to share):**
124 - From OUTSIDE the box: `curl --max-time 5 http://<box-public-ip>:8000/api/health` ?
**refused / timeout** (the engine is `127.0.0.1`-bound; no inbound port). ? The box is not
directly reachable.
125 - `curl -i https://api.khelsutra.guru/api/health` **without** a CF session ? a **302 to
the Access login** (not a 200). ? The gate is in front.
126 - Open `https://app.khelsutra.guru` in a browser ? CF Access login ? after sign-in, the
SPA loads and `/api/health` returns 200 (same session cookie). ?
127
128 **The shareable CUJ (what a tester does):** open `app.khelsutra.guru` ? sign in ?
**upload a video from their device** (the SPA chunked-upload lands it in `upload_dir`, inside
the jail) ? **Process** ? watch progress ? pick rallies ? **Build** ? **download the reel**. All
from the page. *(Pasting a `gs://` URI does NOT work on this box ? see §7: the
`allowed_video_root` jail rejects every non-local URI. Browser upload is the alpha path.)*
129
130 ---
131
132 ## 7. Honest caveats / deferred
133
134 - **`upload_dir` must sit UNDER `allowed_video_root`.** If you enable browser upload, an
upload landing outside the jail is rejected by `/api/process` (a functional break, not a hole).
Keep `upload_dir` inside the jail (§2.2).
135 - **`gs://` / `s3://` / `gdrive://` URIs are REJECTED on the exposed box.** Setting
`allowed_video_root` (required to expose, §2.2) path-jails `/api/process` to *local* paths under
that root; the engine resolves a `gs://` URI to a local path that escapes the jail ? `400`
(pinned by engine `tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`).
Cloud-URI ingest works **only** when `allowed_video_root` is unset (the local single-user
posture ? the opposite of this runbook). For alpha, the ingest path is **browser upload** (it
lands inside the jail). *(Scheme-exempting recognized buckets from the jail would be an engine
change, not a config flip.)*
136 - **You MUST set `edge_auth_enabled=true` to expose.** The boot posture (and the in-app
JWT verify) only engage then. Do not put the engine behind the tunnel with edge auth off.
137 - **Cross-origin reel download:** the reel streams from `api.` with `Content-
Disposition: attachment`, so it downloads even cross-origin ? but verify the filename in your
browser during the CUJ (cross-origin `<a download>` filename hints are ignored; the header is

```

the authority).

```

138     - **Upload size:** Cloudflare's free proxy caps request bodies at **100 MB** ? large
matches upload as ~95 MB chunks (the SPA chunked-upload path honours the engine's `max_part_mb`,
default **95**; lower `security.max_part_mb` to shrink them). `gs://` bucket pre-staging is
**not** an option on this box (see the jail caveat above).
139     - **Box offline = engine offline.** The Pages SPA stays up and shows a friendly "engine
offline" banner (`/api/health` ping); processing resumes when the box is back. Beta removes this
with a queue (`HOSTING_PLAN.md` §3).
140     - **Gemini = paid + uploads footage to Google.** Keep the key rotated; use it only for
clips you're OK sending.
141
142     ---
143
144     ## 8. Teardown
145     `cloudflared` tunnel cleanup khelsutra-studio` + delete the tunnel; remove the Pages
custom domain; disable the Access application; stop the engine service. The box's `output/` +
`uploads/` retain originals ? sweep them per your retention policy before any wider exposure.
146
147     ---
148     ### Changelog
149     - 2026-06-22 ? **correctness pass** (adversarial verify against engine `origin/master` +
a hermetic repro). Fixed two would-break-the-deploy gaps: (1) **`RALLY_ALLOWED_HOSTS` was
missing** ? with `edge_auth` on, the Host-header guard 400s every tunneled request until it's
set (added it + `RALLY_BIND_HOST` to §2.2; repro confirmed 400?200); (2) **`gs://` ingest is
rejected** by the `allowed_video_root` jail on the exposed box (dropped the `gs://` promise from
§6/§7; alpha path = browser upload). Minor: 401-not-500 wording (§2.1), `config.json`
location/`--setup` note (§2.2), bind comment (§2.3), ~95 MB chunk size (§7).
150     - 2026-06-21 ? created. Split-topology alpha deploy kit (cloudflared + Pages + CF
Access) consuming the engine exposure-safety boot posture (engine #294) + the SPA
`VITE_API_BASE` (#64). Pairs with `LOCAL_APPLIANCE_ARCHITECTURE.md` (single-origin alternative)
+ the engine `HOSTING_PLAN.md` §2.
151

```

5. assistant (2026-06-22T09:46:33.089Z)

I have the audit findings for `deploy/DEPLOY\_ALPHA.md`. There are two stale findings. Let me re-verify each against the live repos before editing.

Finding 1: header should be LIVE (line 3, currently `READY`).

Finding 2: host list too narrow (line 66) ? `engine.env.example` adds loopback; suggestedFix "add localhost".

Let me verify both against the live repo evidence.

6. user (2026-06-22T09:46:34.411Z)

```

95
96     **The shareable CUJ:** open `app.khelsutra.guru` ? sign in (email OTP) ? **upload a
video** (lands in the
97     jail) ? Process ? pick rallies ? Build ? download the reel. *(Pasting a `gs://` URI is
rejected by the
98     jail on the exposed box ? upload is the alpha path; see `../DEPLOY_ALPHA.md` §7.)*
99
100    > **? Live-verified on the droplet (2026-06-22).** Engine installed + booted on
`168.144.126.176`; full
101    > suite **1313 passed** + ruff/mypy clean on the box; boot-posture fail-fast, the Host-
guard `400?200`,
102    > the path-jail (`gs://`/`/etc/passwd`/`..` ? 400), the T4 upload?jail, and the edge-
auth `401`-gate all
103    > exercised live. Tunnel + Access + Pages are up; the full **DNS ? Cloudflare ? Access ?
tunnel ? engine**
104    > chain is confirmed (`/api/health` = 200 through an Access session; unauthenticated ?
302 to the login).
105    > Two things still need a human: the rotated **Gemini key** (`/etc/khelsutra/engine.env`
? restart) and a

```

```

106 > **browser one-time-PIN login** to exercise the upload?reel CUJ.
107 >
108 > **? Two gotchas the live run surfaced:**
109 > - **`indexing.default_segmenter` MUST be `gemini` on a CPU box** (set in
`config.json.example`). The SPA
110 > sends no segmenter, so the engine uses its default ? the shipped default
`yolo_hybrid` needs torch/GPU
111 > and every job fails `No module named 'torch'` without this.
112 > - **Without a valid `GEMINI_API_KEY`, a job finishes `done` with 0 rallies** ? the
missing-key error lands
113 > only in the engine log, not the job status. Set the key before sharing.
114

```

7. user (2026-06-22T09:46:34.521Z)

```

1 # KhelSutra engine ? environment for the alpha droplet (the `api.` side).
2 # Copy to /etc/khelsutra/engine.env, fill the secret, and `chmod 600` it.
3 # Loaded by khelsutra-engine.service. Verified against engine origin/master
4 # (see ../DEPLOY_ALPHA.md §2 + khelsutra PR #78).
5
6 # --- Secret (rotate the chat-shared one; Gemini is paid + uploads footage to Google)
7
8 GEMINI_API_KEY=PASTE_YOUR_ROTATED_KEY_HERE
9
10 # --- App home: config.json, .env, the SQLite DB and output/ all resolve UNDER here
(engine P0a).
11 # The engine reads $RALLY_APP_HOME/config.json ? so the security block below lives
at
12 # /srv/khelsutra/config.json (NOT the repo root). Use `--setup` so writer/reader
agree. ---
13 RALLY_APP_HOME=/srv/khelsutra
14
15 # --- Exposure: the two vars the original kit MISSED. Without them the tunnel is dead-
on-arrival. ---
16 # edge_auth_enabled=true flips the bind to 0.0.0.0 AND keeps the Host-header guard
loopback-only,
17 # so it boots "healthy" then 400s every request cloudflared forwards with Host:
api.khelsutra.guru.
18 RALLY_ALLOWED_HOSTS=api.khelsutra.guru,localhost,127.0.0.1 # REQUIRED ? load-bearing
fix (allowlist is decoupled from the bind); loopback added for on-box health checks
19 RALLY_BIND_HOST=127.0.0.1 # keep the loopback bind
(ccloudflared dials 127.0.0.1:8000 anyway)
20 RALLY_CORS_ALLOW_ORIGINS=https://app.khelsutra.guru # lock CORS to the SPA origin

```

8. assistant (2026-06-22T09:46:43.051Z)

Both findings confirmed against the live repo.

Finding 1: `droplet/README.md:100` shows the deploy is "Live-verified on the droplet (2026-06-22)" ? DNS?Cloudflare?Access?tunnel?engine chain confirmed, Pages/Tunnel/Access up. The header on line 3 still says `READY` ("owner executes the cloud steps"), which now lags reality. Confirmed stale.

Finding 2: `engine.env.example:17` sets
`RALLY\_ALLOWED\_HOSTS=api.khelsutra.guru,localhost,127.0.0.1` with the rationale "loopback added for on-box health checks". The DEPLOY_ALPHA.md §2.2 (line 66) only sets
`RALLY\_ALLOWED\_HOSTS="api.khelsutra.guru"`. The actual deployed kit adds the loopback hosts so the on-box `/api/health` check works. Confirmed stale/narrow.

Let me verify the engine's Host-guard behavior to make sure the loopback addition is correct (the `/api/health` posture check in §6 hits via the tunnel with `Host: api.khelsutra.guru`, but on-box checks would use `localhost`/`127.0.0.1`).

9. user (2026-06-22T09:46:43.851Z)

```
origin/master:backend/config/startup.py:141:    # loopback-only when RALLY_ALLOWED_HOSTS is
unset (see main._allowed_hosts_from_env) ? it would
origin/master:backend/config/startup.py:143:    if not (env.get("RALLY_ALLOWED_HOSTS", "") or
"".strip()):
origin/master:backend/config/startup.py:146:                                "security.edge_auth_enabled is ON but
RALLY_ALLOWED_HOSTS is unset ? the Host-header guard "
origin/master:backend/config/startup.py:147:                                "defaults to loopback-only and will 400
forwarded requests. Set RALLY_ALLOWED_HOSTS to your "
origin/master:backend/main.py:33:from starlette.middleware.trustedhost import
TrustedHostMiddleware
origin/master:backend/main.py:89:def _allowed_hosts_from_env(env: Optional[Dict[str, str]] =
None) -> List[str]:
origin/master:backend/main.py:93:    Host headers by default). ``RALLY_ALLOWED_HOSTS`` (comma-
separated) overrides; the default keeps
origin/master:backend/main.py:98:    raw = (src.get("RALLY_ALLOWED_HOSTS", "") or "").strip()
origin/master:backend/main.py:100:    # IPv4 loopback (the default bind) + the TestClient host.
NB: Starlette's TrustedHostMiddleware
origin/master:backend/main.py:110:    TrustedHostMiddleware,
origin/master:backend/main.py:111:    allowed_hosts=_allowed_hosts_from_env(),
origin/master:backend/main.py:1088:    # the Host allowlist (the default; don't set
RALLY_ALLOWED_HOSTS to a non-loopback-only set
origin/master:backend/main.py:1101:                                f"RALLY_ALLOWED_HOSTS to your public
host(s) (the Host-header guard defaults to "
```

10. user (2026-06-22T09:46:46.557Z)

```
def _allowed_hosts_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
    """P0c: Host-header allowlist ? a DNS-rebinding guard for the localhost appliance. A
malicious
    web page can point a domain it controls at 127.0.0.1 and script requests to the local
server;
    the browser sends ``Host: attacker.com``, which this rejects (the server only answers
loopback
    Host headers by default). ``RALLY_ALLOWED_HOSTS`` (comma-separated) overrides; the default
keeps
    Starlette's TestClient host ``testserver`` so the in-process suite isn't broken. Read from
the
    ENV (not config.json) so importing ``backend.main`` does zero filesystem I/O; a
routable/edge
    deploy sets its public host (or ``*`` when fronted by an auth gate that validates the
host)."""
    src = os.environ if env is None else env
    raw = (src.get("RALLY_ALLOWED_HOSTS", "") or "").strip()
    hosts = [h.strip() for h in raw.split(",") if h.strip()]
    # IPv4 loopback (the default bind) + the TestClient host. NB: Starlette's
TrustedHostMiddleware
    # matches on host.split(":")[0], which mangles a bracketed IPv6 Host ("::1:port"), so IPv6
    # loopback is NOT supported by this guard ? access the appliance via 127.0.0.1/localhost.
    return hosts or ["localhost", "127.0.0.1", "testserver"]
```

DNS-rebinding guard. Bound to localhost by default, so only loopback Host headers are allowed (a remote page can't read responses cross-origin AND its Host header is rejected here) so an unexpected ``www.`` host is rejected rather than 307-redirected.

```
app.add_middleware(
    TrustedHostMiddleware,
    allowed_hosts=_allowed_hosts_from_env(),
    www_redirect=False,
)
```

```
def _resolve_bind_host(config: Optional[AppConfig] = None, cli_host: Optional[str] = None) ->
str:
    """P0c: the server bind address. Precedence: ``--host`` > ``RALLY_BIND_HOST`` env >
``0.0.0.0``
    when an edge-auth profile is active (``security.edge_auth_enabled`` ? i.e. there IS an auth
gate
    in front) > ``127.0.0.1`` (default: localhost-only, today's behaviour). Binding a routable
interface with no auth gate is unsafe, hence the localhost default; widen it consciously."""
```

11. assistant (2026-06-22T09:46:52.196Z)

Confirmed. When `RALLY\_ALLOWED\_HOSTS` is set, it REPLACES the default (it does not append loopback). So setting it to only `api.khelsutra.guru` drops `localhost`/`127.0.0.1` from the allowlist ? which would 400 on-box `/api/health` checks against loopback. The deployed kit (`engine.env.example:17`) correctly adds `localhost,127.0.0.1` for on-box health checks. The DEPLOY_ALPHA.md §2.2 value at line 66 is too narrow.

Both findings are verified. Now applying the surgical edits.

****Edit 1: Header status `READY` ? `LIVE`**** (line 3).

12. user (2026-06-22T09:46:54.555Z)

The file C:\Users\avidu\Projects\khelsutra\deploy\DEPLOY_ALPHA.md has been updated successfully. (file state is current in your context ? no need to Read it back)

13. assistant (2026-06-22T09:46:57.220Z)

****Edit 2: Add loopback hosts to `RALLY\_ALLOWED\_HOSTS`**** (line 66), matching the deployed `engine.env.example:17`.

14. user (2026-06-22T09:47:02.549Z)

The file C:\Users\avidu\Projects\khelsutra\deploy\DEPLOY_ALPHA.md has been updated successfully. (file state is current in your context ? no need to Read it back)

15. assistant (2026-06-22T09:47:06.177Z)

Both edits applied. Let me verify the §2.2 ? callout (line 70) doesn't contradict the new value ? it references setting `RALLY\_ALLOWED\_HOSTS="api.khelsutra.guru"` inline as the load-bearing fix.

16. user (2026-06-22T09:47:06.828Z)

70 > ****?** Two env vars you MUST set, or the tunnel is dead-on-arrival.****** With `edge\_auth\_enabled: true`, the engine (a) flips its bind to `0.0.0.0` and (b) keeps the DNS-rebinding ****Host-header guard**** at its loopback-only default ? so it boots "healthy" but then ****400`s** every request `cloudflared` forwards with `Host: api.khelsutra.guru`**\*\*** (including the §6 `/api/health` check). The boot only *\*warns\** about this (`EXPOSED_HOST_ALLOWLIST_LOOPBACK`), it does not stop. Set **\*\*`RALLY_ALLOWED_HOSTS="api.khelsutra.guru"`\*\*** (the load-bearing fix ? the Host allowlist is independent of the bind) and **\*\*`RALLY_BIND_HOST="127.0.0.1"`\*\*** (so the bind stays loopback as §0/§6 claim; `cloudflared` dials loopback anyway). *\*Verified against engine `origin/master`:* with the allowlist unset, `GET /api/health` with `Host: api.khelsutra.guru` ? **\*\*400`**; with it set ? ****200`**.*

17. assistant (2026-06-22T09:47:10.730Z)

Line 70's ? callout still shows the narrow `RALLY\_ALLOWED\_HOSTS="api.khelsutra.guru"` inline, which now disagrees with the §2.2 code block I just fixed. To keep the doc internally consistent and match the deployed `engine.env.example:17`, I'll align this inline reference too. This is within the scope of the "host list too narrow" finding (same value, same section).

18. user (2026-06-22T09:47:13.625Z)

The file C:\Users\avidu\Projects\khelsutra\deploy\DEPLOY_ALPHA.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

19. assistant (2026-06-22T09:47:16.779Z)

All edits applied. The two audited findings are resolved, plus I kept the §2.2 ? callout consistent with the same fix. No code changes were required, and no findings were skipped. Now returning the structured report.

20. user (2026-06-22T09:47:31.718Z)

Structured output provided successfully

21. assistant (2026-06-22T09:47:37.733Z)

I've completed the staleness fix for `C:\Users\avidu\Projects\khelsutra\deploy\DEPLOY\_ALPHA.md`. Both audited findings were re-verified against the live repos and applied surgically:

1. **Header `READY` ? `LIVE`** (line 3) ? confirmed by `deploy/droplet/README.md:100-106` (live-verified on the droplet 2026-06-22; full DNS?Cloudflare?Access?tunnel?engine chain confirmed). Kept honest by noting the rotated Gemini key + browser OTP login still remain to exercise the upload?reel CUJ.

2. **`RALLY\_ALLOWED\_HOSTS` widened** to `api.khelsutra.guru,localhost,127.0.0.1` (line 66) to match the deployed `deploy/droplet/engine.env.example:17`. Verified the rationale against engine `origin/master` `backend/main.py::\_allowed\_hosts\_from\_env` ? setting this var *replaces* the loopback-only default, so the narrow value would 400 on-box health checks. Also aligned the §2.2 ? callout (line 70) with the same value for internal consistency.

No code changes were needed and nothing was skipped.